

PROJECT PROPOSAL

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Abstract

The project aims to tackle a very important accessibility gap: day to day communication between deaf ASL users and hearing non-signers often depends on human interpreters or tools that are slow, costly and unreliable in casual contexts. The objective is to develop a real-time ASL-to-English translation system for mobile devices, that converts visual signs into English text or speech through computer vision and multimodal learning. The solution will integrate image and video recognition (CNNs/Transformers), sequence modeling, and model quantization with a Flutter application.

The project will be organized in phases: clarifying needs and scope, outlining the overall design, building a functional prototype, and validating it with targeted tests. The expected result is a demonstrable prototype that runs in a controlled environment, along with an assessment of its accuracy, responsiveness on a typical smartphone, and brief user feedback to guide future improvements.

Keywords: American Sign Language (ASL), Sign Language Recognition, Real-Time Translation, Computer Vision, Multimodal Learning, Model Quantization.

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1 Project Description

1.1 Introduction

Human-computer interaction for accessibility seeks to lower communication barriers between people with different communication modalities. In everyday life, interactions between Deaf American Sign Language (ASL) users and hearing non-signers often depend on human interpreters or tools that are slow or impractical. This project will address the gap by proposing a real-time ASL-to-English translation system on a mobile device, emphasizing model design and robustness.

The proposed solution will decompose the task into specialized modules: (1) Video preprocessing, including frame extraction and selection of relevant frames; (2) a computer-vision regression model to localize the signing hand by predicting the four corner points of the hand region per frame, enabling perspective-aware cropping and normalization; (3) dynamic sign recognition to model motion across short clips; (4) temporal fusion and decoding will stabilize the frame outputs into sign decisions; (5) language generation, a multimodal vision-language model (PaliGemma [1]) will convert visual sequences (extracted frames and/or keypoints) into fluent English text, with optional text to speech for a spoken output; and (6) real-time optimization, exploring the fine tuning and quantization of the models, to fit the system on a smartphone.

1.2 Objective

The objective of this project will be to design, implement, and validate a real-time ASL-to-English translation system on a mobile device by creating a sign-to-language pipeline with on-device inference [2]. The work will prioritize technical performance (accuracy, stability and latency). Here are the specific goals:

- **Video Preprocessing:** Extract frames from the device camera stream and select relevant frames for downstream processing.
- **Hand Localization (Regression):** Predict the four corner points of the hand region per frame to enable perspective-aware cropping and normalization.
- **Dynamic Sign Recognition:** Model short motion patterns over localized clips to identify a defined vocabulary of isolated signs.
- **Temporal Fusion and Decoding:** Stabilize frame/clip predictions into consistent sign decisions with low jitter.
- **Language Generation (PaliGemma):** Convert visual sequences (frame embeddings and/or keypoints) into fluent English text; enable optional text to speech.

- **On-Device Optimization:** Apply fine tuning and quantization to meet real-time latency and memory constraints on a mid-range smartphone.
- **Mobile Prototype (Flutter):** Implement a mobile application that captures video, runs the pipeline locally, and displays/speaks the output.

1.3 Technical Requirements

To develop the project, it will be necessary to:

- **Video ingestion and preprocessing:** Capture from the device camera, extract frames, and select relevant frames.
- **Hand-region localization:** Train a regression model to predict the four corner points of the hand region and integrate it into the pipeline.
- **Isolated sign recognition and decoding:** Recognize short motion sequences and add a simple temporal fusion layer to stabilize outputs.
- **Multimodal vision–language model:** Use PaliGemma to render English text from visual inputs; integrate optional text to speech.
- **Mobile client:** Implement a Flutter app that runs on-device inference and presents spoken/written output.
- **Real-time optimization:** Apply lightweight optimizations (e.g., quantization / hardware acceleration) to meet latency/memory constraints on a mid-range phone.
- **Evaluation and deliverables:** Specify accuracy/latency metrics and provide a functional prototype operating in a test environment.

1.4 Use Case

This section presents a use case illustrating how the system will operate on a mobile device. The user will open the application and point the camera at the signer. The application will capture video frames, select relevant frames, and localize the hand region. Short clips from the localized region will then be analyzed to recognize isolated signs. Finally, a multimodal vision–language model (PaliGemma) will render the recognized sign(s) as English text, with the option to produce speech output.

When confidence is insufficient, the application will prompt the user to reframe the hand or improve lighting and will defer the final output until the prediction stabilizes. The recognized text (and, if enabled, spoken output) will be presented on screen for immediate use in casual interactions. See Figure 1.

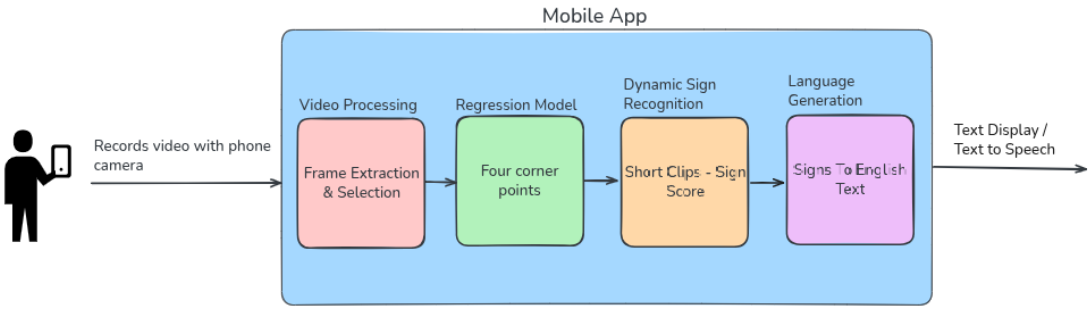


Figure 1: Use case data flow on device

1.5 System Architecture

The project will follow a dual system architecture (Figure 2). The **Client (mobile app)** takes the video made by the user and does a frame extraction, which are then analyzed by the **Hand Recognition Model**, the frames with the hands detected and quad points are then turned into a **SuperImage** of frames, which is sent to a **Server** that orchestrates calls to the **Sign Recognition Service**, **PaliGemma** (language generation), and optional **Text to Speech Service**. The Server returns English text or speech to the Client.

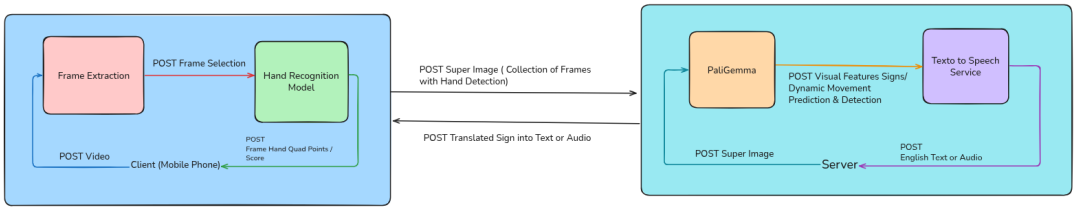


Figure 2: System Architecture

2 Motivation

This project aligns with a strong interest in computer vision and multimodal models, especially the challenge of bridging sign and language for signed input. This will provide me with more practice in model optimization and evaluation metrics, while still trying to produce a complete end-to-end, usable prototype rather than just a theoretical study.

From a professional standpoint, this work will build a strong foundation for my future in which I want to persue a Master's in AI and potential PhD, by combining discipline and experimentation with real-time constraints. The outcome is expected to strengthen my skills in experimental design, quantitative evaluation, and system integration, which are really important competencies in advance studies and research practice.

Although there is no personal connection to the Deaf community, I still feel really attached with this important accessibility problem, and will try by doing technical progress create a tool to reduce everyday communication barriers.

3 Expected Results and Feasibility

This section explains the expected results of the project, the development timeline, the knowledge that I possess so far as well as the one I will be required to get, and the courses involved.

3.1 Expected Results

A mobile application is expected to be delivered that runs the full sign-to-language pipeline on device. The app will capture the video, localize the signing hand through a bounding-box/quadrilateral regression model, and recognize signs using two stages: static hand-shape recognition for letters or fixed poses and dynamic sign recognition for short motion sequence. The recognized signs will then be rendered as English text by a multimodal vision-language model (PaliGemma), with optional text to speech to provide a spoken output. So the final deliverable will include the trained models integrated inside app.

3.2 Estimated Time For Technical Development

The development is divided into components and tasks with start and end dates as show in Table 1. The schedule is subject to changes through out the development of the project.

Component	Tasks	Initial Date	End Date
Project Definition	Previous Research	22/10/2025	27/10/2025
	Project Scope Definition	27/10/2025	31/10/2025
	Project Design	29/10/2025	31/10/2025
Project Development	Hand Localization	29/10/2025	20/11/2025
	Static Handshape Recognition	21/11/2025	10/01/2026
	Dynamic Sign Recognition	11/01/2026	28/02/2026
	Temporal Fusion & Decoding	01/03/2026	14/03/2026
	Language Generation	15/03/2026	14/04/2026
	Mobile Prototype	15/04/2026	08/05/2026
	Component Integration	09/05/2026	16/05/2026
Project Deployment	Deployment	17/05/2026	20/05/2026
	Validation Tests	21/05/2026	30/05/2026
Documentation	Report	01/03/2026	30/05/2026
	Presentation	15/05/2026	30/05/2026
Tutoring Sessions	Weekly meetings	22/10/2025	30/05/2026

Table 1: Project Schedule

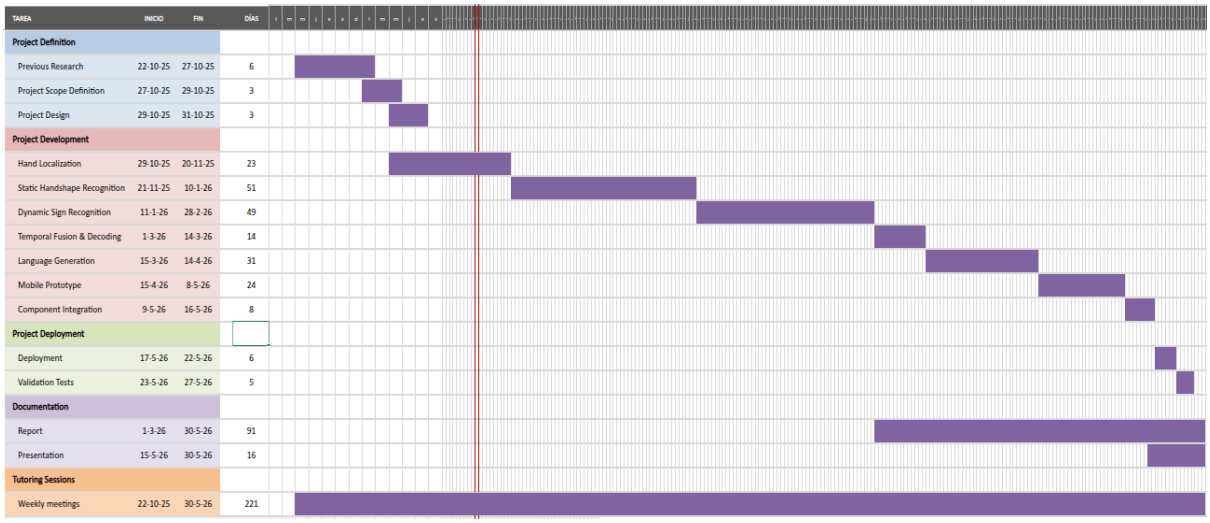


Figure 3: Gantt's Diagram

4 Bibliography

References

[1] L. Beyer, A. Steiner, A. S. Pinto, A. Kolesnikov, X. Wang, D. Salz, M. Neumann, I. Alabdulmohsin, M. Tschannen, E. Bugliarello, T. Unterthiner, D. Keysers, S. Koppula, F. Liu, A. Grycner, A. Gritsenko, N. Houlsby, M. Kumar, K. Rong, J. Eisenschlos, R. Kabra, M. Bauer, M. Bošnjak, X. Chen, M. Minderer, P. Voigtlaender, I. Bica, I. Balazevic, J. Puigcerver, P. Papalampidi, O. Henaff, X. Xiong, R. Soricut, J. Harmsen, and X. Zhai, "Paligemma: A versatile 3b vlm for transfer," *arXiv preprint arXiv:2407.07726*, 2024, version 2, Oct. 10, 2024. [Online]. Available: https://arxiv.org/pdf/2407.07726

[2] J. Lee, N. Chirkov, E. Ignasheva, Y. Pisarchyk, M. Shieh, F. Riccardi, R. Sarokin, A. Kulik, and M. Grundmann, "On-device neural net inference with mobile gpus," *arXiv preprint arXiv:1907.01989*, 2019. [Online]. Available: https://arxiv.org/pdf/1907.01989